

Example: [The x-15 and the wedge tail] Consider $M=7$ flow over a flat plate at an angle of attack of 10° and a 10° wedge. Compare the C_ℓ coefficients.

a) Flat plate; top: App C for $M=7$: $v_1 = 90.97^\circ$

$$v_2 = v_1 + \alpha = 100.97$$

From tables $M_2 = 9.56$

From isentropic tables

$$p_{t1}/p = 0.414 \times 10^4 ; p_{t2}/p = 0.33 \times 10^5$$

$$\text{Thus } \frac{p_2}{p_1} = \frac{p_{t1}/p}{p_{t2}/p} = \frac{0.414 \times 10^4}{0.33 \times 10^5} = 0.1255$$

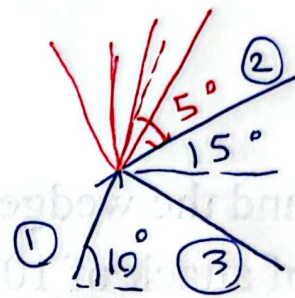
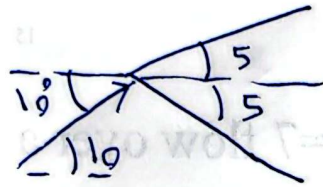
Bottom: from δ - θ - M for $M_1=7$, $\delta=10^\circ$ we get: $\theta = 16.5$

$$M_{n1} = M_1 \sin \theta = 7 \times \sin(16.5) = 1.99$$

From shock tables for $M_{n1} = 1.99$ we get $p_3/p_1 = 4.45$

$$\text{For the } C_\ell \text{ is: } \Rightarrow C_\ell = \frac{2}{\gamma M_1^2} \left(\frac{p_3}{p_1} - \frac{p_2}{p_1} \right) \cos \alpha = \frac{2}{1.4 \times 7^2} (4.45 - 0.1255) = 0.126$$

b)



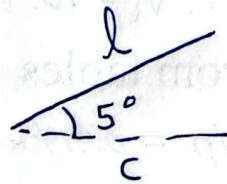
①-② Exp. Wave

turn angle 5°

$$v_1 = \checkmark \rightarrow v_2 = v_1 + 5$$

$$\xrightarrow[\text{Calc}]{\text{online}} M_2 = 8.1, \frac{P_{t2}}{P_2} = 1.0897 \times 10^4$$

From geometry



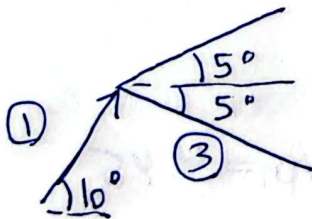
$$l \cos 5^\circ = c \rightarrow l = \frac{c}{\cos 5^\circ}$$

Force on surface 2

$$F_2 = P_2 l = \frac{P_{t2}}{P_1} \times P_1 l = 0.38 P_1 l$$

$\frac{P_{t2}}{P_2} = 1.0897 \times 10^4$

①-③

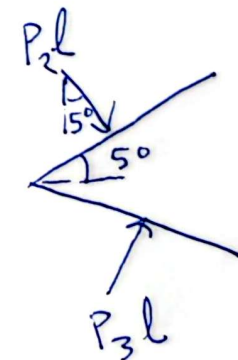


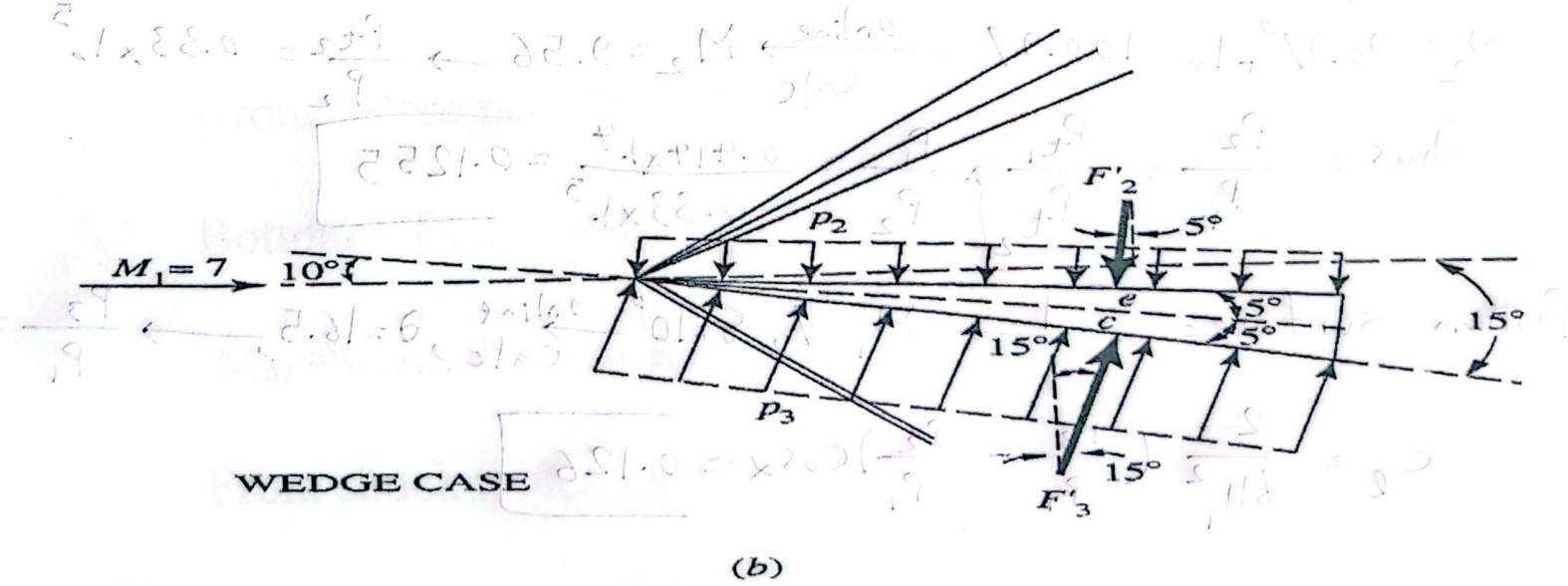
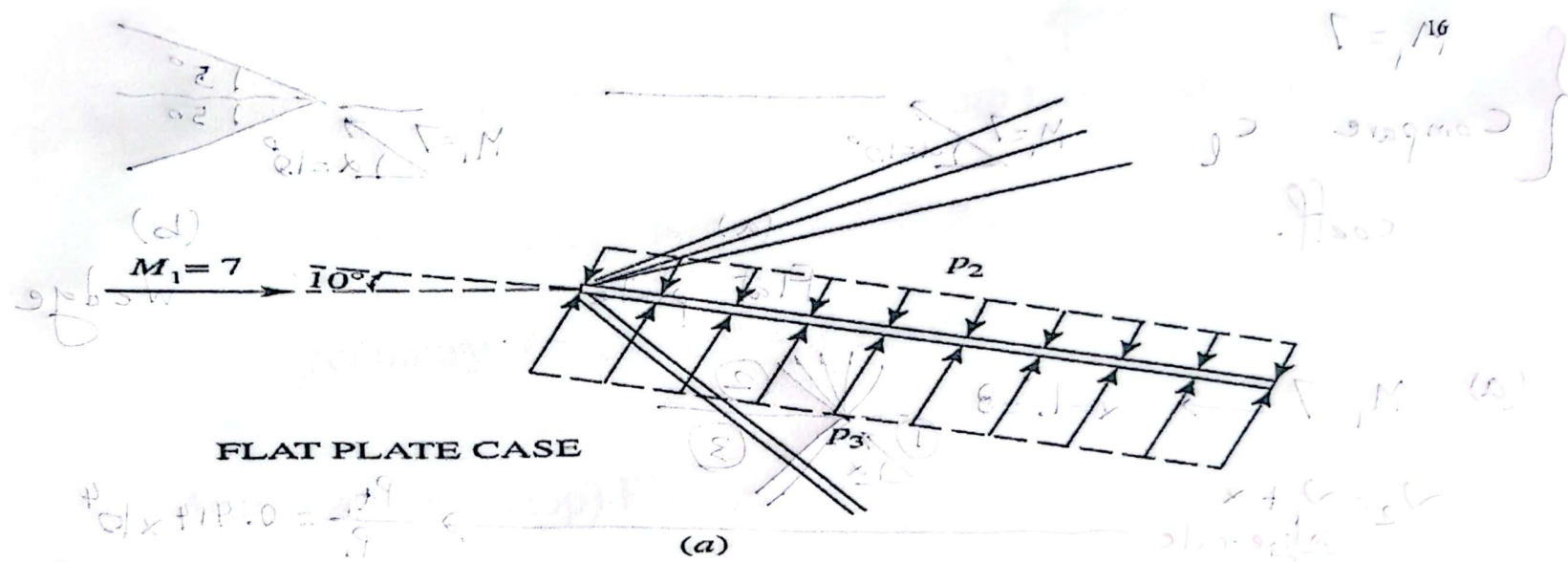
①-③

$$\left\{ \begin{array}{l} M_1 = 7, \delta = 15^\circ \\ \frac{P_3}{P_1} = 8.915 \end{array} \right. \xrightarrow[\text{Calc}]{\text{online}} \theta = 23.5^\circ$$

$$F_3 = P_3 l = \left(\frac{P_3}{P_1} \right) P_1 l = 8.915 P_1 l$$

$$L = F_3 \cos 5^\circ - F_2 \cos 5^\circ = 8.232 P_1 l = 8.265 P_1 e$$



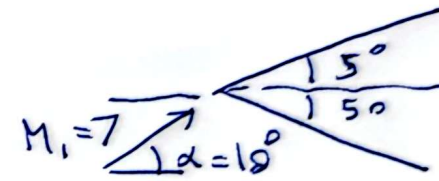


Note: Angles not to scale

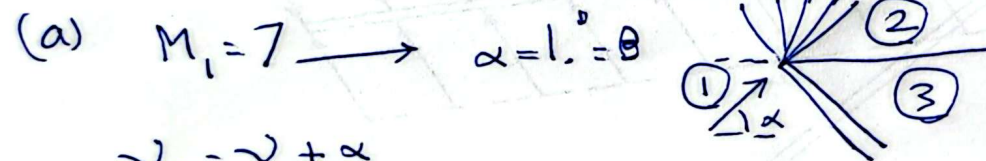
$M_1 = 7$
 { Compare c_l
 coeff.



(a)
Flat plate



(b)
wedge



$$\nu_2 = \nu_1 + \alpha \xrightarrow{\text{online-calc}} \frac{P_{t1}}{P_1} = 0.414 \times 10^4$$

$$\nu_2 = 90.97^\circ + 10 = 100.97 \xrightarrow{\text{online Calc}} M_2 = 9.56 \rightarrow \frac{P_{t2}}{P_2} = 0.33 \times 10^5$$

$$\text{Thus: } \frac{P_2}{P_1} = \frac{P_{t1}}{P_{t2}} \times \frac{P_1}{P_2} = \frac{0.414 \times 10^4}{0.33 \times 10^5} = 0.1255$$

Bottom surface: $\xrightarrow{1} M_1 = 7, \delta = 10^\circ \xrightarrow{\text{online Calc}} \theta = 16.5, \rightarrow \frac{P_3}{P_1} = 4.45$

$$c_l = \frac{2}{\gamma M_1^2} \left(\frac{P_3}{P_1} - \frac{P_2}{P_1} \right) \cos \alpha = 0.126$$

b) Wedge; top: $v_2 = v_1 + \frac{S}{(10-5)}$; From tables $M_2 = 8.1$ ^{expansion}

From isentropic tables $p_2/p_1 = 1.0897 \times 10^{-4}$

From geometry $l = c / \cos 5^\circ = 1.0096$

$$\text{Force/unit span (top)} F_2 = p_2 l = \left(\frac{p_2/p_1}{p_2/p_1} \right) p_1 l = \left(\frac{0.914 \times 10^{-4}}{1.0897 \times 10^{-4}} \right) p_1 l = 0.838 p_1 l$$

(from before $p_2/p_1 = 0.914 \times 10^{-4}$)

Bottom: from δ - θ - M for $M_1 = 7$, $\delta = 15^\circ$ we get: $\theta = 23.5^\circ$

$$M_{n1} = M_1 \sin \theta = 7 \times \sin(23.5) = 2.79$$

From shock tables for $M_{n1} = 2.79$ we get $p_3/p_1 = 0.915$

Thus $F'_3 = p_3 \ell = (p_3/p_1)p_1 \ell = 8.915 p_1 \ell$

Lift: $L' = F'_3 \cos 15^\circ - F'_2 \cos 5^\circ = (0.9659)(8.915)p_1 \ell$
 $- (0.9962)(0.38)p_1 \ell$
 $= 8.232 p_1 \ell = 8.265 p_1 c$

($\ell = 1.004c$)

Lift coefficient: $C_\ell = \frac{L'}{(\gamma/2)p_1 M_1^2 c} = \frac{8.265 p_1 c}{(\frac{1.4}{2})p_1 7^2 c} = 0.241$

Higher for wedge. Why? bottom surface has inclination 15° vs. $10^\circ \rightarrow$

Stronger shock $\rightarrow$ higher $\frac{p_3}{p_1} \rightarrow$

(8.915 vs 4.45)

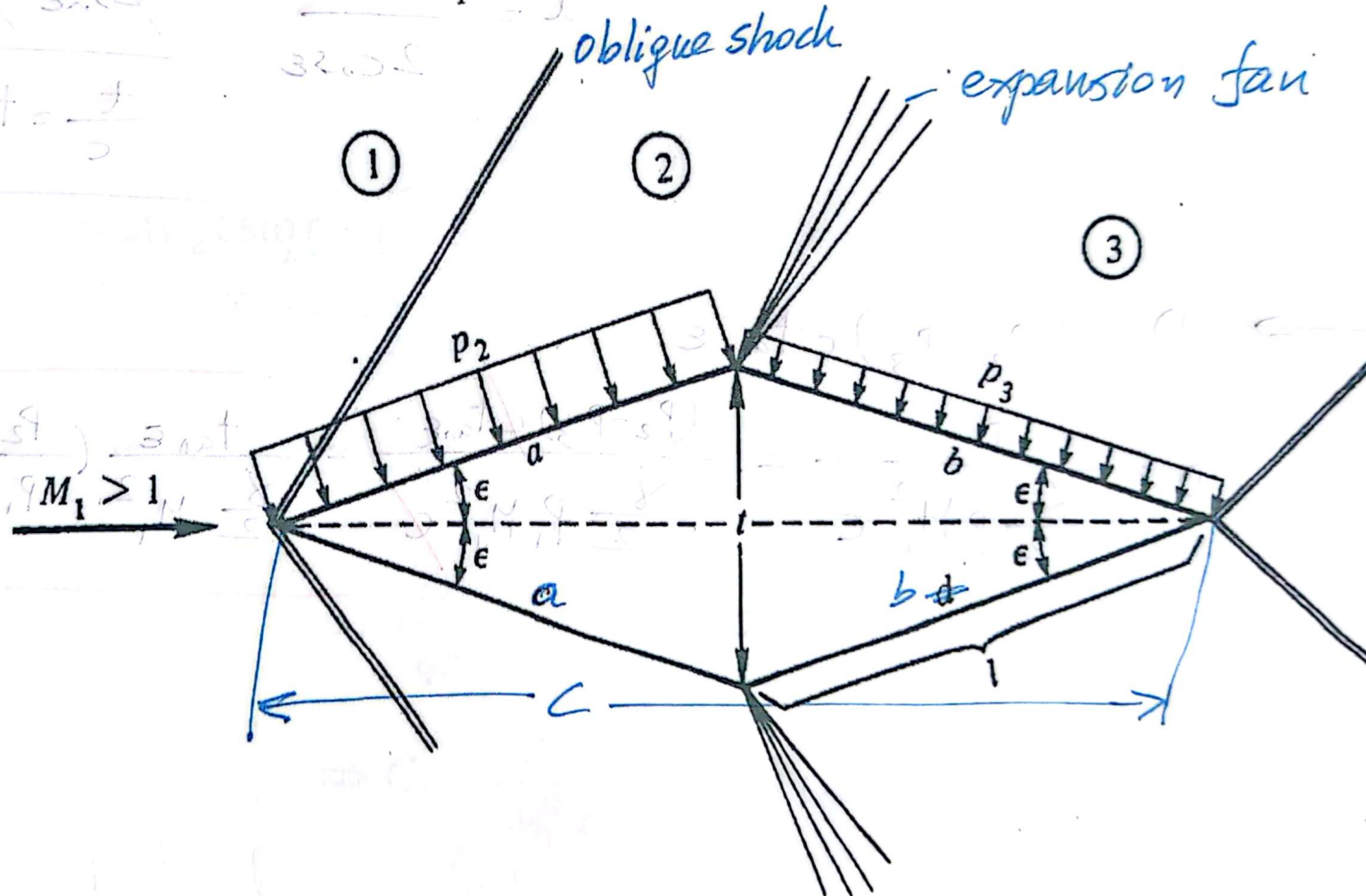
higher lift

(expansion on the top has small effect)

$$C_L = \frac{L}{\frac{\delta}{2} P_1 M_1^2} = 0.241 > \underbrace{0.126}_{\text{Flat plate}}$$

Diamond airfoil: $M_1 > 1$

$\alpha = 0$



$$l = \frac{t}{2 \sin \epsilon}$$

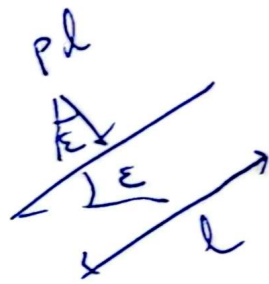
$$l = \frac{c}{2 \cos \epsilon}$$

$$\Rightarrow \frac{t}{2 \sin \epsilon} = \frac{c}{2 \cos \epsilon}$$

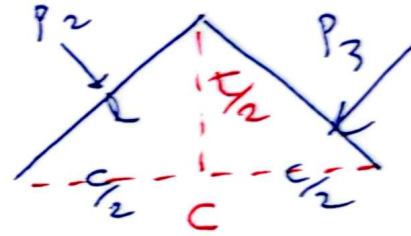
$$\boxed{\frac{t}{c} = \tan \epsilon}$$

$$\Rightarrow D = (P_2 - P_3) c \tan \epsilon$$

$$C_D = \frac{D}{\frac{\gamma}{2} P_1 M_1^2 c} = \frac{(P_2 - P_3) \cancel{c} \tan \epsilon}{\frac{\gamma}{2} P_1 M_1^2 \cancel{c}} = \frac{\tan \epsilon}{\frac{\gamma}{2} M_1^2} \left(\frac{P_2}{P_1} - \frac{P_3}{P_1} \right)$$



$$p l \sin \epsilon$$



20

drag/unit depth

propulsive force

$$D' = 2(p_2 l \sin \epsilon - p_3 l \sin \epsilon) = (p_2 - p_3) t$$

to get to horizontal

upper + lower surface

$$l \sin \epsilon = \frac{t}{2}, l \cos \epsilon = \frac{c}{2}$$

(shock wave drag)

$$t = 2l \sin \epsilon$$

$$D = (p_2 - p_3) t$$

$$l = \frac{c}{2 \cos \epsilon}$$

$$C_d = \frac{D}{q c} \quad \left(\frac{l}{c} = \frac{1}{2 \cos \epsilon} \right)$$

$$\Rightarrow C_d = \frac{2}{\gamma M_1^2} \frac{t}{c} \left(\frac{p_2}{p_1} - \frac{p_3}{p_1} \right) \neq 0$$

$$p_2 > p_1$$

oblique shock

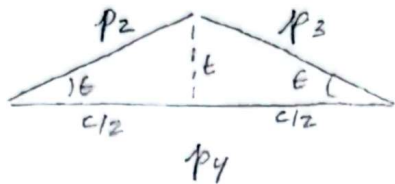
$$p_3 < p_1$$

expansion fan

$$C_l = 0 \quad (\alpha = 0; \text{symmetry})$$

shock wave drag

How to find the force coefficients



$$\tan \epsilon = \frac{t}{c/2} = \frac{2t}{c}$$

$$t = \frac{c}{2} \tan \epsilon$$

($b = \text{span}$)

Normal force:

$$F_N = \left(p_4 c - p_2 \frac{c}{2} - p_3 \frac{c}{2} \right) b$$

$$F_N = \left(2 \frac{p_4}{p_1} - \frac{p_2}{p_1} - \frac{p_3}{p_1} \right) \frac{bc}{2} p_1$$

$$C_N = \frac{F_N}{\frac{1}{2} \rho \infty V_\infty^2 bc} = \frac{F_N}{\frac{\gamma p_1}{2} M_\infty^2 bc}$$

$$C_N = \left(2 \frac{p_4}{p_1} - \frac{p_2}{p_1} - \frac{p_3}{p_1} \right) \frac{bc}{2} p_1 \frac{2}{\gamma p_1 M_\infty^2 bc}$$

$$C_N = \frac{1}{\gamma M_\infty^2} \left(2 \frac{p_4}{p_1} - \frac{p_2}{p_1} - \frac{p_3}{p_1} \right)$$

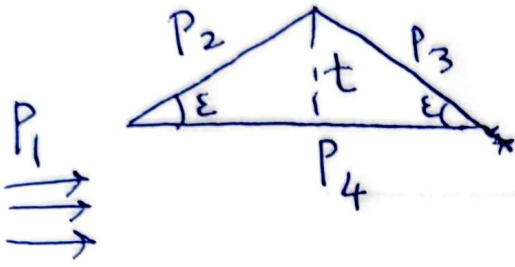
Axial Force:

$$F_A = p_2 t b - p_3 t b = (p_2 - p_3) \frac{cb}{2} \tan \epsilon$$

$$C_A = \frac{2 F_A}{\gamma p_1 M_\infty^2 bc} = \frac{2}{\gamma p_1 M_\infty^2 bc} \frac{bc}{2} \tan \epsilon (p_2 - p_3)$$

$$C_A = \frac{1}{\gamma M_\infty^2} \left(\frac{p_2}{p_1} - \frac{p_3}{p_1} \right) \tan \epsilon$$

"Triangular airfoils"



$$\tan \epsilon = \frac{t}{c/2} = \frac{2t}{c}$$

$$t = \frac{c}{2} \tan \epsilon$$

$$F_N = P_4 c - P_2 \frac{c}{2} - P_3 \frac{c}{2}$$

$$F_N = \left(2 \frac{P_4}{P_1} - \frac{P_2}{P_1} - \frac{P_3}{P_1} \right) \frac{c}{2} P_1$$

$$C_N = \frac{F_N}{\frac{\rho}{2} P_1 M_1^2 c} = \frac{1}{\gamma M_1^2} \left(2 \frac{P_4}{P_1} - \frac{P_2}{P_1} - \frac{P_3}{P_1} \right)$$

Note: Normal force or lift is geometry independent in final relation (No ϵ appears) but depend on pressure on surfaces

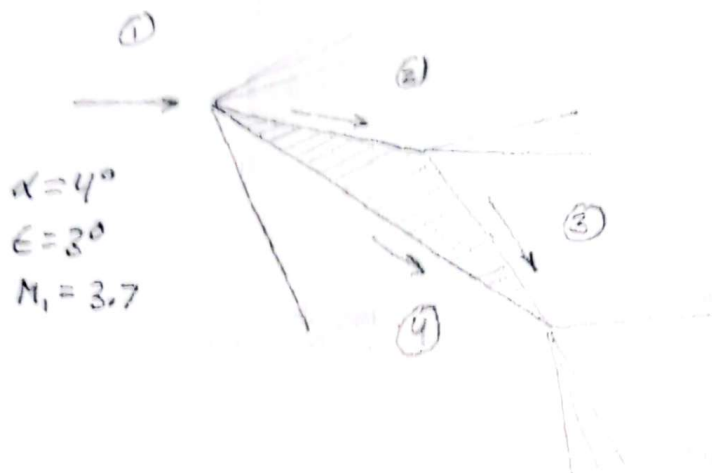
Axial force: Surface 4 does not contribute in drag as Force is in normal direction on this surface.

$$F_A = P_2 t - P_3 t = (P_2 - P_3) \frac{c}{2} \tan \epsilon$$

$$C_A = \frac{F_A}{\frac{\rho}{2} P_1 M_1^2 c} = \frac{1}{\gamma M_1^2} \left(\frac{P_2}{P_1} - \frac{P_3}{P_1} \right) \tan \epsilon$$

Drag is geometry-dependent (ϵ appears)

2.)



① → ② expansion

$$\theta_{12} = \alpha - \epsilon = 1^\circ$$

$$\nu(M_2) = \nu(M_1) + \theta_{12} = 61.60^\circ + 1^\circ = 62.60^\circ$$

$$M_2 = 3.769$$

$$\frac{p_2}{p_1} = \frac{p_2/p_0}{p_1/p_0} = \frac{9.008 \times 10^{-3}}{9.903 \times 10^{-3}}$$

$$\boxed{\frac{p_2}{p_1} = 0.9096}$$

② → ③ expansion

$$\theta_{23} = 2\epsilon = 6^\circ$$

$$\nu(M_3) = \nu(M_2) + \theta_{23} = 62.60^\circ + 6^\circ = 68.60^\circ$$

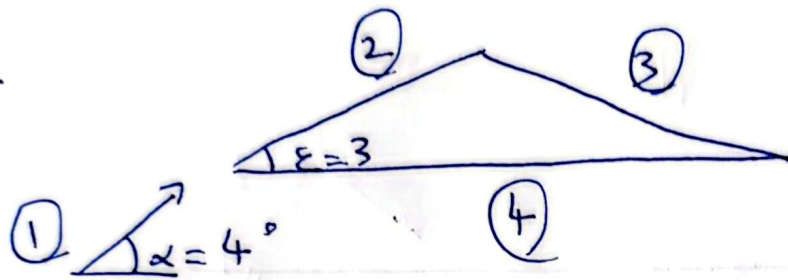
$$M_3 = 4.221$$

$$\frac{p_3}{p_1} = \frac{p_3/p_0}{p_1/p_0} = \frac{4.924 \times 10^{-3}}{9.903 \times 10^{-3}} \Rightarrow \boxed{\frac{p_3}{p_1} = 0.4972}$$

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(Flow on top of airfoil is all isentropic)

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E_x.



$$M_1 = 3.7$$

$$\textcircled{1}-\textcircled{2} \quad E.W \quad \delta = 1^\circ$$

$$\textcircled{2}-\textcircled{3} \quad E.W \quad \delta = 6^\circ$$

$$\textcircled{1}-\textcircled{4} \quad O.S \quad \delta = 4^\circ$$

$$\textcircled{1}-\textcircled{2} \quad \nu(M_2) = \nu(M_1) + \delta = 61.60 + 1 = 62.60$$

$$M_2 = 3.769, \quad \frac{P_2}{P_1} = \frac{P_2/P_t}{P_1/P_t} = 0.9096$$

$$\textcircled{2}-\textcircled{3} \quad E.W \rightarrow \nu(M_3) = \nu(M_2) + 6^\circ = 62.60 + 6^\circ = 68.60^\circ \rightarrow M_3 = 4.221$$

$$\frac{P_3}{P_1} = \frac{P_3/P_0}{P_1/P_0} = 0.4972$$

$$\textcircled{1}-\textcircled{4} \rightarrow O.S \rightarrow \begin{cases} \delta = 4^\circ \\ M_1 = 3.7 \end{cases} \rightarrow \begin{cases} \theta = 18.48^\circ \\ M_4 = 3.439 \end{cases}$$

$$\frac{P_4}{P_1} = 1.438$$

$$C_N = \frac{1}{8M_1^2} \left(2 \frac{P_4}{P_1} - \frac{P_2}{P_1} - \frac{P_3}{P_1} \right)$$

$$= \frac{1}{1.4 \times 3.7^2} (2 \times 1.438 - 0.9096 - 0.4972) = \underline{7.666 \times 10^{-2}}$$

$$C_A = \frac{1}{8M_1^2} \left(\frac{P_2}{P_1} - \frac{P_3}{P_1} \right) \tan \epsilon$$

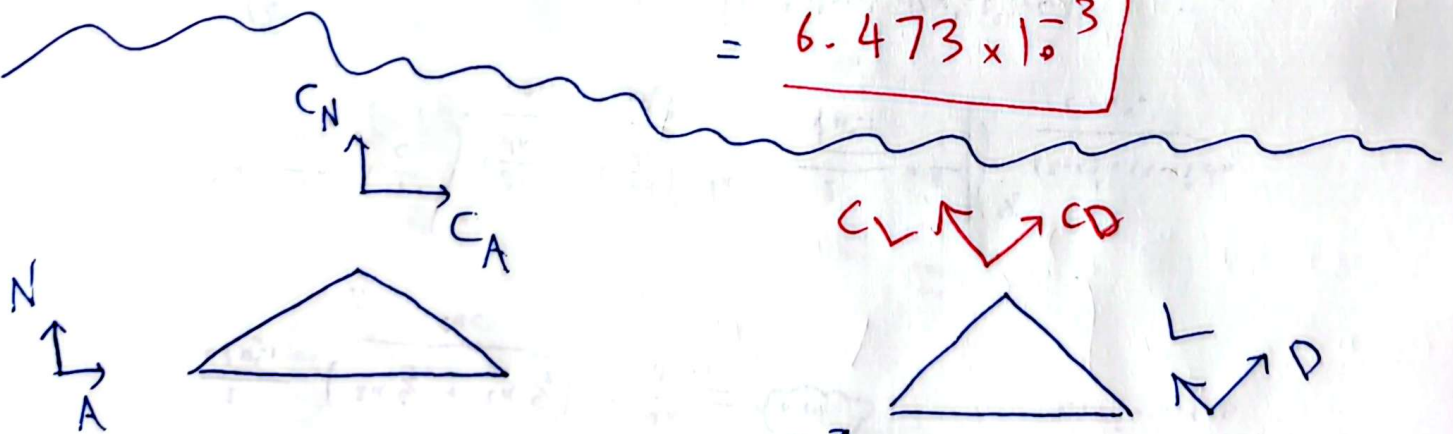
$$= \frac{1}{1.4 \times 3.7^2} [0.9096 - 0.4972] \tan 3^\circ = \underline{1.128 \times 10^{-3}}$$

$$C_L = C_N \cos \alpha - C_A \sin \alpha = 7.66 \times 10^{-2} \cos 4^\circ - 1.128 \times 10^{-3} \sin 4^\circ$$

$$= 7.640 \times 10^{-2} \approx C_N$$

$$C_d = C_N \sin \alpha + C_A \cos \alpha = 7.66 \times 10^{-2} \sin 4^\circ + 1.128 \times 10^{-3} \cos 4^\circ$$

$$= \underline{6.473 \times 10^{-3}}$$



$C_L \nearrow \searrow C_D$

$\Delta \alpha$

$D \parallel \alpha$

$L \perp \alpha$

Linear Theory [10.1]

Starting with Euler's equation

$$dp = -\rho U dU \Rightarrow p - p_\infty = -\rho U (U - U_\infty)$$

Then 8.57 becomes:

$$\frac{dU}{U} = \frac{d\theta}{\sqrt{M_\infty^2 - 1}} \Rightarrow \frac{U - U_\infty}{U} = \frac{\theta}{\sqrt{M_\infty^2 - 1}}$$

$$\Rightarrow C_p = \frac{2\theta}{\sqrt{M_\infty^2 - 1}} \quad \text{rads} \quad (10.1)$$

Sign of θ needs to be consistent; also

$$\theta_u = \frac{dz_u}{dx} - \alpha \quad \text{upper}$$

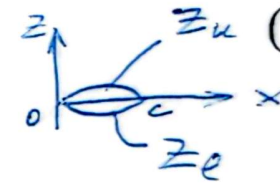
$$\theta_l = -\frac{dz_l}{dx} + \alpha \quad \text{lower} \quad \text{wrt } V_\infty \quad (10.2)$$

Lift

$$dl = p_l ds_l \cos \theta_l - p_u ds_u \cos \theta_u \Rightarrow dl = (p_l - p_u) dx \quad (10.4)$$

Nondimensionalizing: $dC_l = (C_{pl} - C_{pu}) d\left(\frac{x}{c}\right)$ (10.5)

$$dC_l = \frac{2}{\sqrt{M_\infty^2 - 1}} \left(2\alpha - \frac{dz_l}{dx} - \frac{dz_u}{dx} \right) d\left(\frac{x}{c}\right) \quad (10.6)$$



note: $\int_0^c \left(\frac{dz_u}{dx} + \frac{dz_l}{dx} \right) dx = 0$

Integrating:

$$C_l = \frac{4\alpha}{\sqrt{M_\infty^2 - 1}} \quad \text{rads}$$

→ Geometry-independent!
(10.8)
Just depends on α

Lift curve slope:

$$\frac{dC_l}{d\alpha} = \frac{4}{\sqrt{M_\infty^2 - 1}} \quad (10.9)$$

Drag

(dd is a better notation, but we follow the text)

$$dd = p_l ds_l \sin \theta_l - p_u ds_u \sin \theta_u \quad (10.10)$$

$\sin \theta \approx \theta$

Small disturbances: $dd = p_l ds_l \theta_l - p_u ds_u \theta_u$ *($ds_l = ds_u \approx dx$)* (10.11)

Nondimensionalizing: *note: $\frac{p}{p_\infty} \theta / \frac{1}{2} \gamma M_\infty^2 = (\frac{p}{p_\infty} - 1) \theta / \frac{1}{2} \gamma M_\infty^2 + \theta / \frac{1}{2} \gamma M_\infty^2$*

$$dC_d = (C_{pl} \theta_l - C_{pu} \theta_u) d\left(\frac{x}{c}\right) + \frac{2}{\gamma M_\infty^2} (\theta_l - \theta_u) d\left(\frac{x}{c}\right) \quad (10.12)$$

$$dC_d = \frac{2}{\sqrt{M_\infty^2 - 1}} \left[2\alpha^2 + \left(\frac{dz_u}{dx}\right)^2 + \left(\frac{dz_l}{dx}\right)^2 \right] d\left(\frac{x}{c}\right) + \left[\frac{-4\alpha}{\sqrt{M_\infty^2 - 1}} \left(\frac{dz_u}{dx} + \frac{dz_l}{dx}\right) + \frac{2}{\gamma M_\infty^2} \left(\frac{dz_u}{dx} - \frac{dz_l}{dx}\right) \right] d\left(\frac{x}{c}\right) \quad (10.13)$$

Integrating:

$$C_d = \frac{4\alpha^2}{\sqrt{M_\infty^2 - 1}} + \frac{2}{\sqrt{M_\infty^2 - 1}} \int_0^1 \left[\left(\frac{dz_u}{dx} \right)^2 + \left(\frac{dz_l}{dx} \right)^2 \right] d\left(\frac{x}{c}\right) \quad (10.14)$$

For small angles

$$\frac{dz_u}{dx} = \tan \sigma_u \approx \sigma_u$$

Define: $\frac{1}{c} \int_0^c \left[\left(\frac{dz_u}{dx} \right)^2 \right] dx \equiv \bar{\sigma}_u^2$; $\frac{1}{c} \int_0^c \left[\left(\frac{dz_l}{dx} \right)^2 \right] dx \equiv \bar{\sigma}_l^2$ (10.15)

$d = d'$ (drag force per unit span)

$$C_d = \frac{d}{q_\infty c} = \frac{4}{\sqrt{M_\infty^2 - 1}} \left[\alpha^2 + \frac{1}{2} (\bar{\sigma}_u^2 + \bar{\sigma}_l^2) \right] \quad (10.16)$$

An alternative formula using camber and thickness:

$$C_d = \frac{4}{\sqrt{M_\infty^2 - 1}} (\alpha^2 + \bar{\sigma}_c^2 + \bar{\sigma}_t^2)$$

Where: $z_c = (z_u + z_l) / 2$; $z_t = (z_u - z_l) / 2$
camber *thickness*

$$\frac{1}{c} \int_0^c \left[\left(\frac{dz_c}{dx} \right)^2 \right] dx \equiv \bar{\sigma}_c^2 ; \quad \frac{1}{c} \int_0^c \left[\left(\frac{dz_t}{dx} \right)^2 \right] dx \equiv \bar{\sigma}_t^2$$

In general: $C_d = C_{d, \text{due to lift}} + C_{d, \text{due to shape}} + C_{d, \text{viscous}}$ (10.17)
+ induced drag (3D)

Pitch Moment

$$dm_{x0} = (p_u - p_l)(x - x_o) dx \quad (10.18)$$

Nondimensionalizing:

$$dC_{m_{xo}} = (C_{pu} - C_{pl}) \left(\frac{x - x_o}{c} \right) d \left(\frac{x}{c} \right) \quad (10.19)$$

using (10.1) $dC_{m_{xo}} = \frac{2}{\sqrt{M_\infty^2 - 1}} (\theta_u - \theta_l) \left(\frac{x - x_o}{c} \right) d \left(\frac{x}{c} \right) \quad (10.20)$

$C_p = \frac{2\theta}{\sqrt{M_\infty^2 - 1}}$ $C_{m_{xo}} = \frac{4}{\sqrt{M_\infty^2 - 1}} \left[\alpha \left(\frac{x_o}{c} - \frac{1}{2} \right) + \bar{m}_{c_{xo}} \right] \quad (10.22)$

$$\bar{m}_{c_{xo}} = \int_0^1 \frac{dz_c}{dx} \left(\frac{x - x_o}{c} \right) d \left(\frac{x}{c} \right)$$

For leading edge moment:

$$C_{m\ le} = \frac{4\alpha}{\sqrt{M_\infty^2 - 1}} \left(-\frac{1}{2} + \bar{m}_{cle} \right)$$

Center of pressure:

effect of camber

$$\frac{x_{cp}}{c} = -\frac{C_{m\ l.e.}}{C_n} = \frac{(2\alpha - 4\bar{m}_{cle}) / \sqrt{M_\infty^2 - 1}}{4\alpha / \sqrt{M_\infty^2 - 1}} = \frac{1}{2} - \frac{\bar{m}_{cle}}{\alpha}$$

Aerodynamic Center:

need to cancel to be independent of α

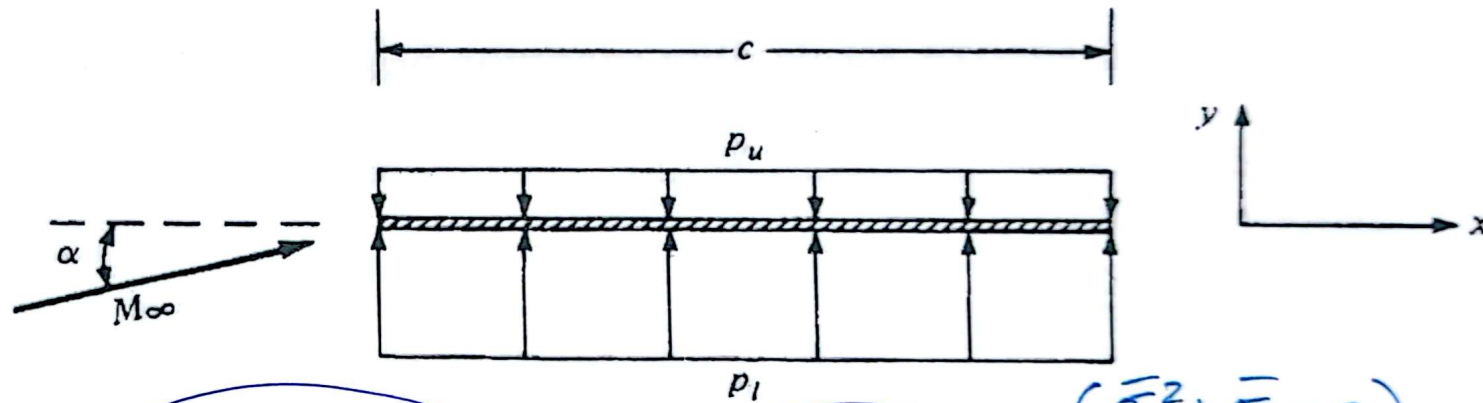
$$C_{m\ a.c.} = C_{m\ l.e.} + \frac{x_{ac}}{c} C_l = \frac{1}{\sqrt{M_\infty^2 - 1}} \left(-2\alpha + 4\bar{m}_{cle} + \frac{x_{ac}}{c} 4\alpha \right)$$

Therefore: $x_{ac} = c/2$ and $C_{m\ a.c.} = \frac{4\bar{m}_{cle}}{\sqrt{M_\infty^2 - 1}}$

(for incompressible: $x_{ac} = \frac{c}{4}$)

Examples

Flat plate at a small angle of attack α



linearized
theory

$$C_l = \frac{4\alpha}{\sqrt{M_\infty^2 - 1}}$$

$$C_d = \frac{4\alpha^2}{\sqrt{M_\infty^2 - 1}}$$

$(\bar{\sigma}_c^2 + \bar{\sigma}_t = 0)$

$$C_{m \text{ l.e.}} = -\frac{2\alpha}{\sqrt{M_\infty^2 - 1}}$$

Center of pressure: $\frac{x_{cp}}{c} = \frac{1}{2}$ (no camber)

Example

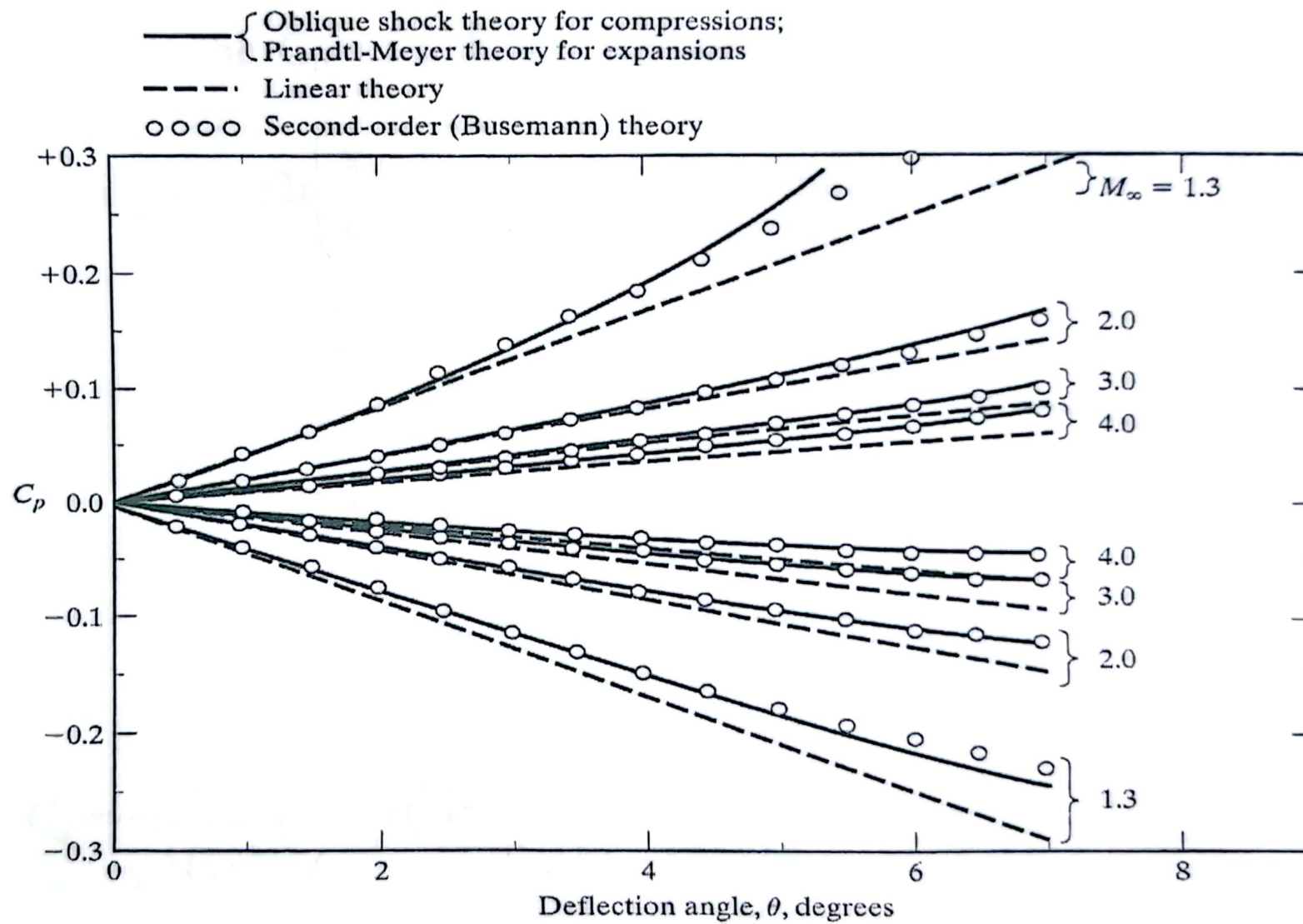
Using linearized theory find C_ℓ , C_d for a flat plate at $M_\infty = 2.6$ at $\alpha = 5^\circ, 15^\circ, 30^\circ$ and compare with shock expansion theory

	<u>linear theory</u>	<u>exact theory</u> (oblique shock, expansion fan)	<u>error</u>
a) $\alpha = 5^\circ$ $C_\ell = \frac{4\alpha}{\sqrt{M_\infty^2 - 1}} = 0.1455$		0.148	1.69%
$C_d = \frac{4\alpha^2}{\sqrt{M_\infty^2 - 1}} = C_\ell \alpha = 0.0127$		0.0129	1.53%

	L	E	33
b) $\alpha = 15^\circ$ $C_\ell =$	0.436	0.452	3.41%
$C_d =$	<u>0.114</u>	0.121	5.7%
c) $\alpha = 30^\circ$ $C_\ell =$	0.873	1.19	26.7%
$C_d =$	<u>0.457</u>	0.687	33.5%

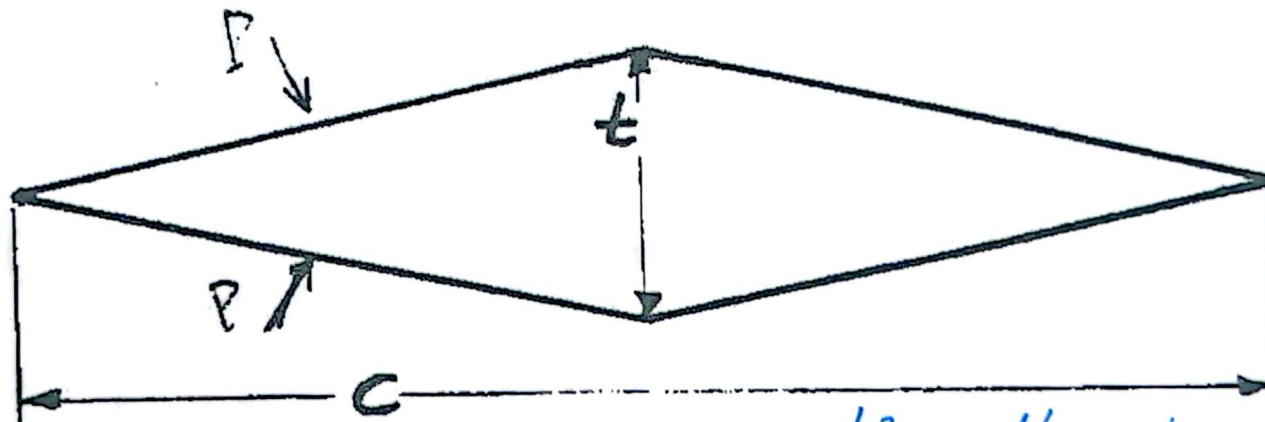
Conclusion: at low α linear theory is good; its accuracy deteriorates with high α , as expected.

[Repeat this example for $M=5$ (validity limit of linearized theory) and check results]



Compare the C_d of a diamond shape and a parabolic (biconvex)

airfoil $\left(Y = \pm 2t \left(\frac{x}{c} - \frac{x^2}{c^2} \right) \right)$ for the same max thickness t at $\alpha = 0$;



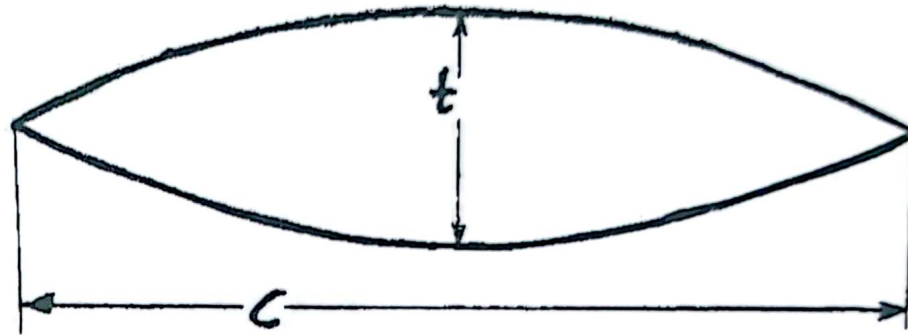
$$C_d = \frac{4}{\sqrt{M_\infty^2 - 1}} (\underbrace{\bar{\sigma}_c^2}_{=0} + \bar{\sigma}_t^2) = \frac{1}{c} \frac{4}{\sqrt{M_\infty^2 - 1}} \int_0^c \underbrace{\left(\frac{dz}{dx} \right)^2}_{\text{thickness}} dx = \frac{4}{\sqrt{M_\infty^2 - 1}} \left(\frac{t}{c} \right)^2$$

$\left| \frac{dz}{dx} \right| = \frac{t/2}{c/2} = \frac{t}{c}$

$\therefore C_d \text{ for diamond airfoil}$

$$M_\infty > 1$$

$$\alpha = 0$$



$$z(x) = \pm 2t \left(\frac{x}{c} - \frac{x^2}{c^2} \right) \Rightarrow \frac{dz}{dx} = \pm \frac{2t}{c} \left(1 - \frac{2x}{c} \right)$$

upper

$$C_d = \frac{1}{c} \frac{4}{\sqrt{M_\infty^2 - 1}} \int_0^c \left(\frac{dz}{dx} \right)^2 dx = \frac{1}{c} \frac{4}{\sqrt{M_\infty^2 - 1}} \int_0^c \frac{4t^2}{c^2} \left(1 - \frac{4x}{c} + \frac{4x^2}{c^2} \right) dx$$

lower

thickness

(camber = 0)

$$= \frac{1}{c} \frac{4}{\sqrt{M_\infty^2 - 1}} \cdot \frac{4t^2}{c^2} \left[c - \frac{4c^2}{2c} + \frac{4}{c^2} \frac{c^3}{3} \right]$$

$$= 1 - 2 + \frac{4}{3} = \frac{1}{3}$$

⇒ parabolic airfoil has 33% more drag at $\alpha = 0$

$$C_d = \frac{4}{\sqrt{M_\infty^2 - 1}} \left(\frac{4}{3} \right) \left(\frac{t}{c} \right)^2$$

at $\alpha \neq 0$?

$$C_\ell = \frac{4\alpha}{\sqrt{M_\infty^2 - 1}}$$

$$C_{d \downarrow \text{diamond}} = \frac{4}{\sqrt{M_\infty^2 - 1}} \left[\alpha^2 + \left(\frac{t}{c} \right)^2 \right]$$

Diamond

$$C_{d \downarrow \text{parabolic}} = \frac{4}{\sqrt{M_\infty^2 - 1}} \left[\alpha^2 + \frac{4}{3} \left(\frac{t}{c} \right)^2 \right]$$

Parabola

effect of thickness

(camber is 0 in both cases)

L/D is independent of M_∞ :

parabolic:

$$\frac{C_\ell}{C_d} = \frac{\alpha}{\alpha^2 + \frac{4}{3} \left(\frac{t}{c} \right)^2}$$

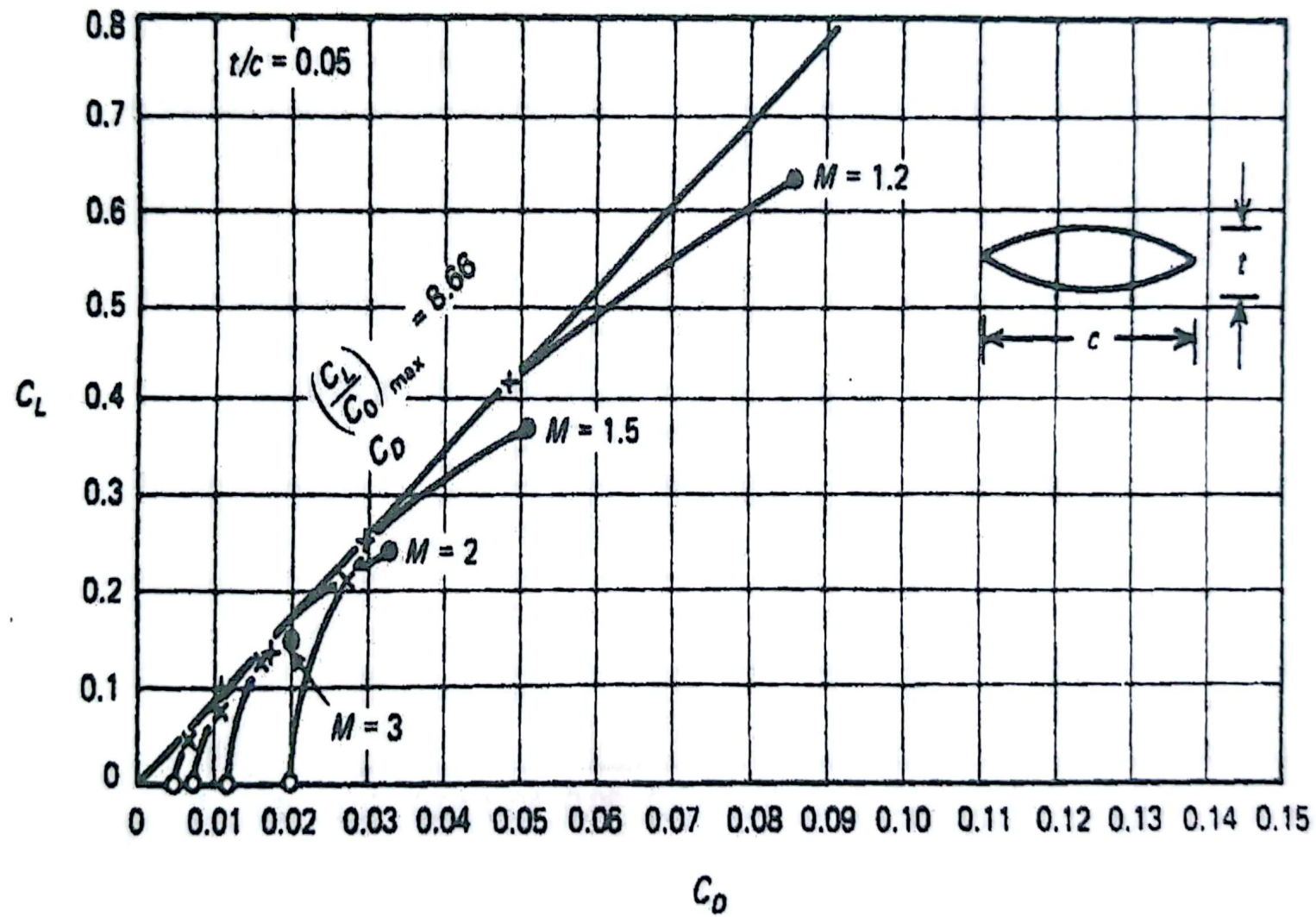
$$\max \frac{C_\ell}{C_d} \text{ when } \alpha = \sqrt{\frac{4}{3}} \frac{t}{c} \Rightarrow \left(\frac{C_\ell}{C_d} \right)_{\max} = \frac{1}{\sqrt{\frac{16}{3}} \left(\frac{t}{c} \right)}$$

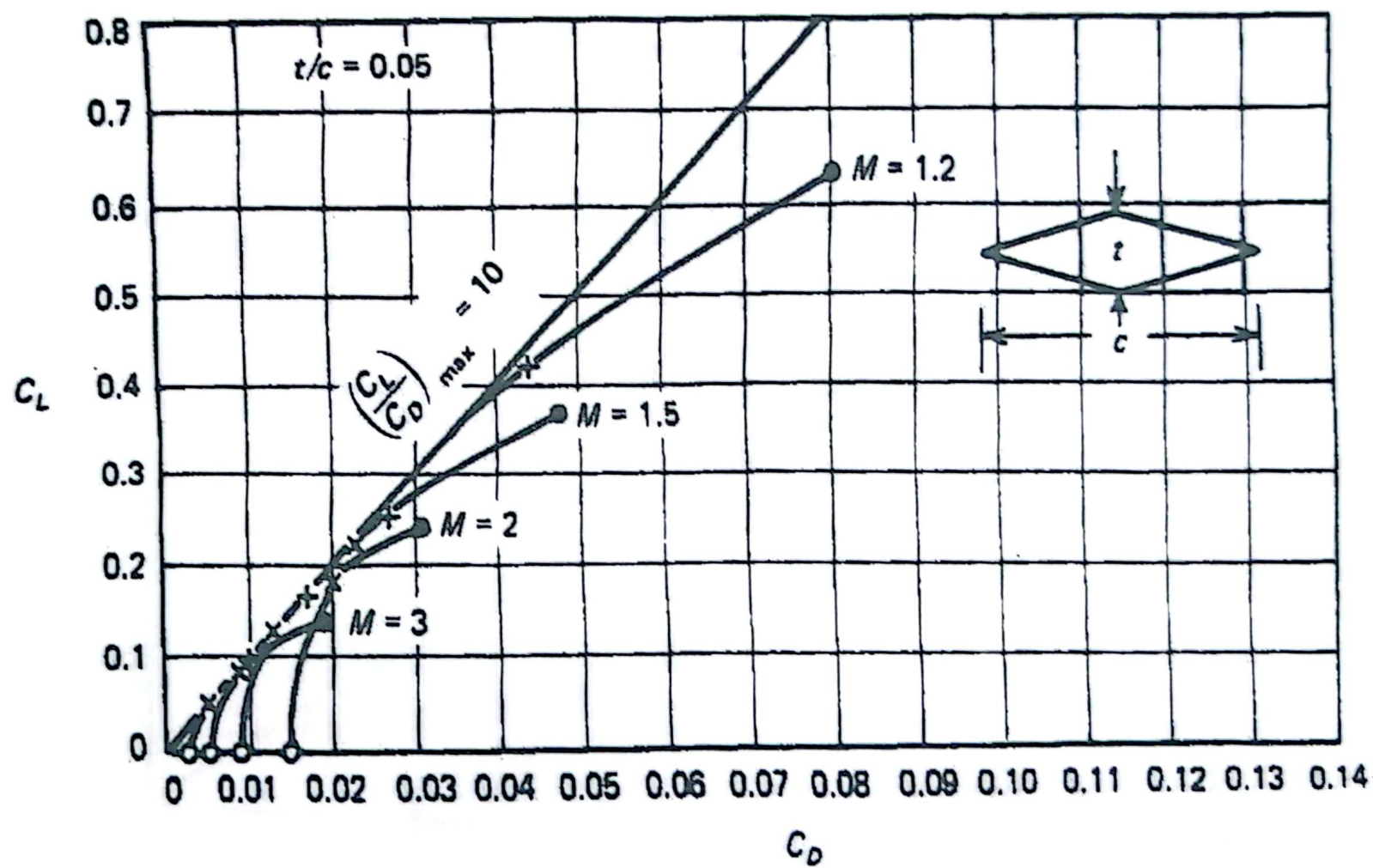
(take $\frac{\partial (C_\ell/C_d)}{\partial \alpha} = 0 \Rightarrow \dots$)

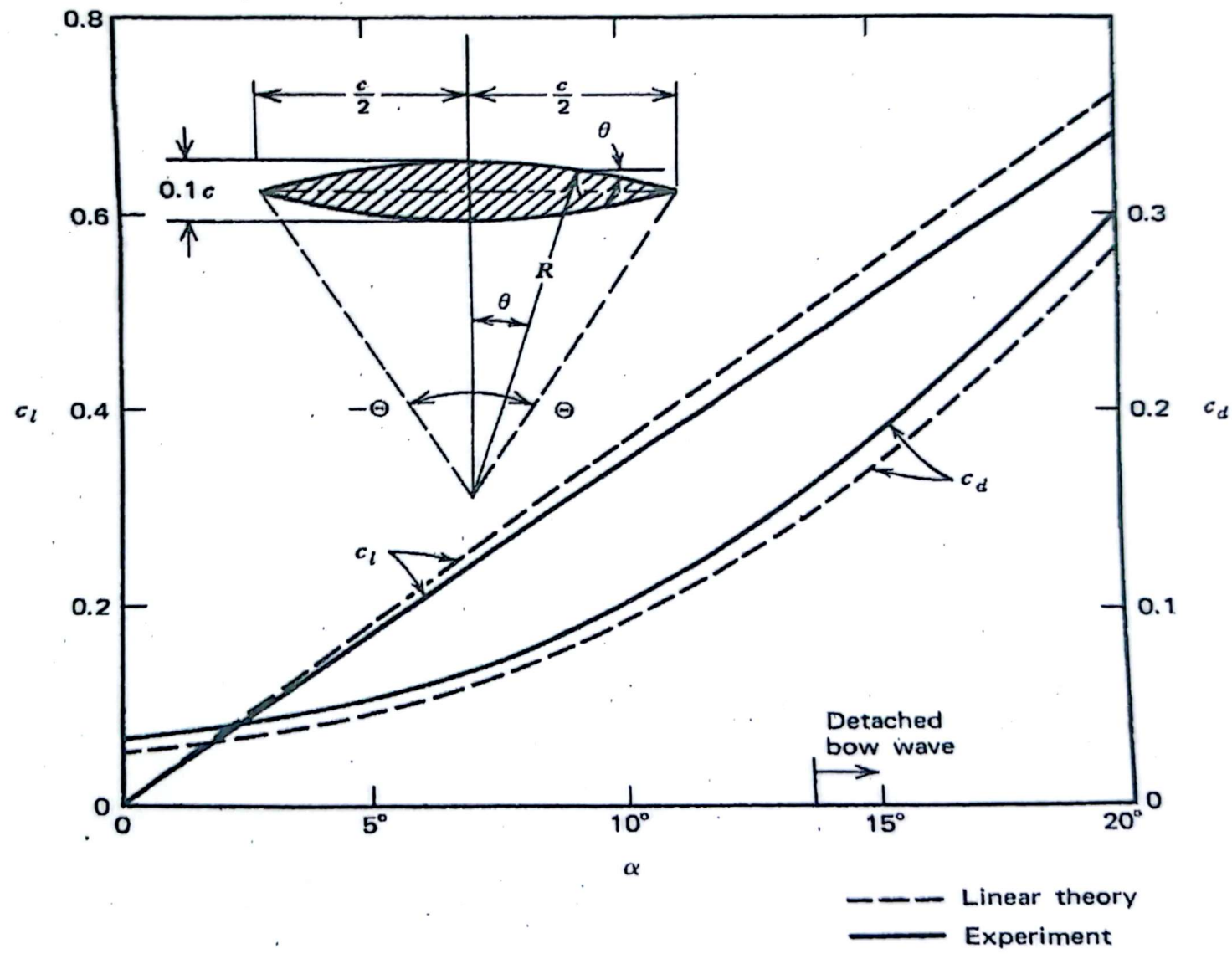
diamond:

$$\frac{C_\ell}{C_d} = \frac{\alpha}{\alpha^2 + \left(\frac{t}{c} \right)^2}$$

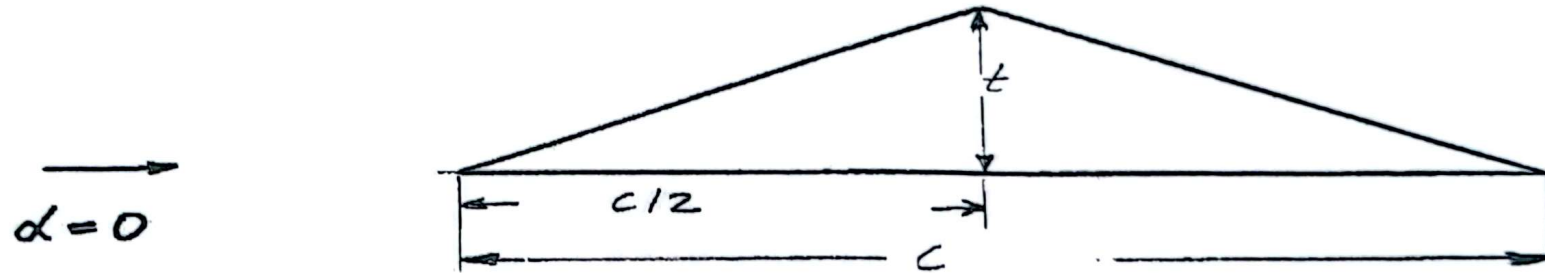
$$\max \frac{C_\ell}{C_d} \text{ when } \alpha = \frac{t}{c} \Rightarrow \left(\frac{C_\ell}{C_d} \right)_{\max} = \frac{1}{2 \left(\frac{t}{c} \right)}$$







Example: triangular airfoil



$$C_d = \frac{2}{\sqrt{M_\infty^2 - 1}} (\bar{\sigma}_u^2 + \bar{\sigma}_t^2); \quad \bar{\sigma}_u^2 = \frac{1}{c} \int_0^c \left(\frac{dz_u}{dx} \right)^2 dx = \frac{1}{c} \left[\int_0^{c/2} \left(\frac{t}{c/2} \right)^2 + \int_{c/2}^c \left(\frac{-t}{c/2} \right)^2 \right] dx$$

$$\bar{\sigma}_t^2 = 0$$

$$= \frac{2}{c} \left[\int_0^{c/2} \left(\frac{t}{c/2} \right)^2 dx \right] = 4 \left(\frac{t}{c} \right)^2$$

" C_d of triangle" $\Rightarrow$ $C_d = \frac{4}{\sqrt{M_\infty^2 - 1}} \left[2 \left(\frac{t}{c} \right)^2 \right]$

$\Rightarrow C_d$ of triangle is higher than the C_d of diamond

note: $c_e = 0$ for $\alpha = 0$ ($c_e = \frac{4\alpha}{\sqrt{M_\infty^2 - 1}}$) although the airfoil is cambered